PATENT APPLICATION ASSIGNMENT

WHEREAS, I, The undersigned, at the indicated addresses given below, respectively, have invented certain new and useful improvements in SOFTWARE-DEFINED OPTICS (SDO) FOR PREDICTIVE MOTION CORRECTION IN MOBILE FREE-SPACE OPTICAL COMMUNICATION (FSOC) LINKS**,** for which an application for U.S. Letters Patent was filed on March 16, 2026, as Serial No. 64/007,864.

WHEREAS, by virtue of my employment/appointment with the UNIVERSITY OF CENTRAL FLORIDA, I am obligated to assign any rights in the said invention to the UNIVERSITY OF CENTRAL FLORIDA or its designee;

WHEREAS, the UNIVERSITY OF CENTRAL FLORIDA has designated the UNIVERSITY OF CENTRAL FLORIDA RESEARCH FOUNDATION, INC., existing by virtue of the laws of the State of Florida, and having an office at 3100 Technology Parkway, Ste. 200, Orlando, FL 32826, to be the assignee of the entire right, title and interest in and to the said invention;

WHEREAS, the UNIVERSITY OF CENTRAL FLORIDA RESEARCH FOUNDATION, INC. is desirous of acquiring the entire right, title and interest in and to said invention and in and to any Letters Patent which may be granted therefor in the United States and in any and all foreign countries;

NOW, THEREFORE, in view of good and valuable consideration, the sufficiency and receipt of which is hereby acknowledged, I, the undersigned, have sold, assigned, and transferred, and by these presents do sell, assign, and transfer, unto said UNIVERSITY OF CENTRAL FLORIDA RESEARCH FOUNDATION, INC., its successors and assigns, the full and exclusive right to the said invention in the United States and its territorial possessions and in all foreign countries and the entire right, title, and interest in and to any and all Letters Patent which may be granted therefor in the United States and its territorial possessions and in any and all foreign countries and in and to any and all divisions, reissues, continuations, and extensions thereof.

I hereby authorize and request the Patent Office Officials in the United States and in any and all foreign countries to issue any and all of said Letters Patent, when granted, to said UNIVERSITY OF CENTRAL FLORIDA RESEARCH FOUNDATION, INC., as the assignee of the entire right, title and interest in and to the same, for the sole use and behalf of said UNIVERSITY OF CENTRAL FLORIDA RESEARCH FOUNDATION, INC., its successors and assigns.

FURTHER, I agree that I will communicate to said UNIVERSITY OF CENTRAL FLORIDA RESEARCH FOUNDATION, INC., or its representatives, any facts known to me respecting said invention; testify in any legal proceedings; sign all lawful papers; execute all divisional, continuation, substitution, renewal, and reissue applications; execute all necessary assignment papers to cause any and all of said Letters Patent to be issued to said UNIVERSITY OF CENTRAL FLORIDA RESEARCH FOUNDATION, INC.; make all rightful oaths; and generally do everything possible to aid the said UNIVERSITY OF CENTRAL FLORIDA RESEARCH FOUNDATION, INC., its successors and assigns, to obtain and enforce proper protection for said invention in the United States and in any and all foreign countries.

IN TESTIMONY WHEREOF, I have hereunto set my hand on the date written herein below.

Application No.: 64/007,864

Date: 3/19/2026 Signed ______________ Signed by: Murat Yuksel B82B9845A35B48B... ______

Murat Yuksel
3100 Technology Parkway, Ste. 200,
Orlando, FL 32826

For an acknowledgment in an individual capacity: (FS 117.05[13][b])
If signature authentication/acknowledgement takes place in another state, territory or district of the United States, cross out "Florida" and insert the applicable state; the requirements of Florida Statute 92.50(2) must be met, and if in a foreign country, the requirements of Florida Statute 92.50(3) must be met.

STATE OF FLORIDA
COUNTY OF __________

The foregoing instrument was acknowledged before me on this ______ day of _________ 20__, by ____________________ (name of person acknowledging.)

(Seal)

____
Signature of Notary Public

____
Print, Type/Stamp Name of Notary

Personally known:_________
OR Produced Identification:__________
Type of Identification Produced:________

Application No.: 64/007,864

Date: 3/19/2026 ____________________ Signed ____________

Signed by:
Saadman Yasar
837BDFF05254466...

Saadman Yasar
3100 Technology Parkway, Ste. 200,
Orlando, FL 32826

For an acknowledgment in an individual capacity: (FS 117.05[13][b])

If signature authentication/acknowledgement takes place in another state, territory or district of the United States, cross out "Florida" and insert the applicable state; the requirements of Florida Statute 92.50(2) must be met, and if in a foreign country, the requirements of Florida Statute 92.50(3) must be met.

STATE OF FLORIDA
COUNTY OF __________

The foregoing instrument was acknowledged before me on this ______ day of _________ 20__, by ____________________ (name of person acknowledging.)

(Seal)

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Signature of Notary Public

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Print, Type/Stamp Name of Notary

Personally known:_________
OR Produced Identification:__________
Type of Identification Produced:________

Application No.: 64/007,864

Date: 3/22/2026

DocuSigned by:
Dr. Svetlana Shtrom
3431C1AE5C15478...

UNIVERSITY OF CENTRAL FLORIDA RESEARCH FOUNDATION, INC.

Name: Svetlana Shtrom

Title: Vice President of Technology Transfer

For an acknowledgment in an individual capacity: (FS 117.05[13][b])

STATE OF FLORIDA
COUNTY OF __________

The foregoing instrument was acknowledged before me on this ____ day of _________, 20__,

By ____________________ (name of person acknowledging.)

(Seal)

____
Signature of Notary Public

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Print, Type/Stamp Name of Notary

Personally known:_________

OR Produced Identification:__________
Type of Identification Produced:________

SOFTWARE-DEFINED OPTICS (SDO) FOR PREDICTIVE MOTION CORRECTION IN MOBILE FREE-SPACE OPTICAL COMMUNICATION (FSOC) LINKS

[001] **FEDERALLY SPONSORED RESEARCH OR DEVELOPMENT**

[002] This invention was made with government support under Award No. 2115215 awarded by the National Science Foundation. The government has certain rights in the invention.

[003] **BACKGROUND OF THE INVENTION**

[004] Laser communication systems offer extremely high bandwidth, license-free operation, strong immunity to electromagnetic interference, and inherent physical-layer security under line-of-sight (LoS) operation. These advantages make laser communication particularly attractive for inter-unmanned aerial vehicle (UAV) communication in next-generation aerial networks. However, the narrow divergence of optical beams makes such links highly sensitive to misalignment caused by UAV vibrations, turbulence, and dynamic platform motion. Even sub-degree pointing deviations can lead to severe coupling losses at the receiver, especially when compact apertures are used. Maintaining reliable optical alignment under such disturbances requires both high-speed actuation and intelligent control. Conventional pointing, acquisition, and tracking (PAT) systems depend on camera or sensor feedback to correct beam drift, but feedback latency and limited update rates restrict their ability to respond to rapid UAV fluctuations. Alternatively, predictive control using data from the onboard inertial measurement unit (IMU) offers the potential for predictive beam steering, thereby anticipating motion before misalignment occurs.

[005] Recent research on UAV-based laser communication systems has primarily focused on improving beam alignment accuracy and adaptive control under platform motion and atmospheric turbulence. Despite these advancements, existing methods remain largely

feedback-driven, reacting to misalignments after they occur. Such reactive control limits response speed and degrades link stability under rapid UAV motion.

[006] However, in view of the art considered as a whole at the time the present invention was made, it was not obvious to those of ordinary skill in the field how reliable optical alignment under UAV motion could be successfully maintained without degrading response speed and link stability.

[007] While certain aspects of conventional technologies have been discussed to facilitate disclosure of the invention, Applicant in no way disclaims these technical aspects, and it is contemplated that the claimed invention may encompass one or more of the conventional technical aspects discussed herein.

[008] The present invention may address one or more of the problems and deficiencies of the prior art discussed above. However, it is contemplated that the invention may prove useful in addressing other problems and deficiencies in a number of technical areas. Therefore, the claimed invention should not necessarily be construed as limited to addressing any of the particular problems or deficiencies discussed herein.

[009] In this specification, where a document, act or item of knowledge is referred to or discussed, this reference or discussion is not an admission that the document, act or item of knowledge or any combination thereof was at the priority date, publicly available, known to the public, part of common general knowledge, or otherwise constitutes prior art under the applicable statutory provisions; or is known to be relevant to an attempt to solve any problem with which this specification is concerned.

[0010] SUMMARY OF THE INVENTION

[0011] In various embodiments, the present invention provides a predictive beam-steering framework based on deep reinforcement learning (DRL). A Double Deep Q-Network (DDQN) predicts unmanned aerial vehicle (UAV) motion from inertial measurement unit (IMU) telemetry and issues predictive correction commands to a micro-electro-mechanical system (MEMS) mirror at the transmitter. The framework is validated through experiments and Zemax OpticStudio (ZOS) simulations linked to UAV motion data, demonstrating marked improvements in beam confinement and coupling efficiency under dynamic conditions.

[0012] In the present invention, a DDQN is developed to predict UAV motion from IMU telemetry and to generate predictive beam-steering commands, enabling predictive optical alignment instead of reactive correction. An integrated control framework combines IMU sensing, DDQN-based prediction, and MEMS mirror actuation through a hybrid onboard–offboard computation pipeline, balancing real-time response with training complexity. A dynamic ZOS model is linked to UAV telemetry via the ZOS API to simulate beam propagation, misalignment effects, and receiver aperture optimization under realistic motion perturbations.

[0013] The proposed framework achieves up to $3\times$ tighter beam confinement and $2.5\times$ higher coupling efficiency than IMU-only control, with 93.5% and 66.2% of beam footprints confined within 20 mm and 75 mm receiver radii, respectively.

[0014] Overall, the present invention provides the first end-to-end integration of DRL and optical modeling for UAV-based optical links, achieving predictive beam stabilization and quantifiable optical performance gains under both static and dynamic flight conditions.

[0015] **BRIEF DESCRIPTION OF THE DRAWINGS**

[0016] FIG. 1 illustrates (a) Overall system model of the adaptive laser communication framework for a UAV, integrating neural prediction, beam steering, and optical transmission subsystems. (b) Hardware design showing the optical path, MEMS mirror, and onboard processing units. (c) ZOS model of the optical link. (Left) Ray-trace diagram with labeled elements. (Right) Three receiver-plane beam footprints from distinct configurations that emulate misalignment due to UAV vibration.

[0017] FIG. 2 illustrates a proposed Double Deep Q-Network (DDQN) architecture for UAV motion prediction and beam correction. The environment generates sequential state–

action–reward–next-state–done ($\langle s, a, r, s', d \rangle$) tuples stored in the replay buffer. The Q-network estimates $Q(s, a; \theta)$ while the target network provides delayed estimates $Q(s', a; \theta_{targ})$ for stable learning. Target values follow the Bellman update, optimized via Adam with soft target updates ($\tau = 10^{-3}$) and ϵ-decay to balance exploration and exploitation.

[0018] FIG. 3 illustrates geometric configuration of the UAV-based laser communication test setup, wherein eight stationary receivers are placed in two elevation layers—ground station (violet) and UAV-altitude (red)—forming a 2×2 grid at each height with spacing $d = 100$ m.

[0019] FIG. 4 illustrates a prediction performance of the DDQN model under UAV hovering and motion. (a) Predicted and true beam footprint trajectories in *X*, *Y*, and *Z* during hovering. (b) Hovering errors with MAE of 5.34, 6.06, and 8.78 mm for *X*, *Y*, and *Z*. (d) Predicted and true trajectories during translational motion. (e) Motion errors with MAE of 88.67, 72.96, and 18.93 mm. (c) and (f) 3D heatmaps illustrate beam footprint density with containment zones at $R = \{10, 20\}$ mm (hovering) and $R = \{25, 75\}$ mm (motion), comparing DDQN correction (green) with baseline IMU-only control (red).

[0020] FIG. 5 is a graphical illustration of coupling efficiency under various prediction horizons. Coupling efficiency results for all eight trials are averaged and plotted in (a) for hovering and (b) for motion, with insets showing logarithmic receiver-radius scaling.

[0021] **DETAILED DESCRIPTION OF THE INVENTION**

[0022] In the following detailed description of the preferred embodiments, reference is made to the accompanying drawings, which form a part thereof, and within which are shown by way of illustration specific embodiments by which the invention may be practiced. It is to be understood that other embodiments may be utilized, and structural changes may be made without departing from the scope of the invention. The invention accordingly comprises the features of construction, combination of elements, and arrangement of parts that will be exemplified in the disclosure set forth hereinafter and the scope of the invention will be indicated in the claims.

[0023] As used in this specification and the appended claims, the singular forms "a," "an," and "the" include plural referents unless the content clearly dictates otherwise. As used in this specification and the appended claims, the term "or" is generally employed in its sense including "and/or" unless the context clearly dictates otherwise.

[0024] Laser-based optical wireless systems have immense potential to increase the capacity and security of Unmanned Aerial Vehicle (UAV) communication. However, UAV-induced vibrations and dynamic instabilities severely degrade optical beam alignment, resulting in significant link losses.

[0025] In various embodiments, the present invention provides a Software-Defined Optics (SDO) framework that integrates machine learning–based predictive control with real-time inertial feedback for resilient UAV laser communications. A Double Deep Q-Network (DDQN) is trained on inertial measurement unit (IMU) telemetry to forecast UAV motion and predictively steer a micro-electro-mechanical system (MEMS)-based laser transmitter, thereby compensating for latency in the sensing–processing–actuation loop. The framework is validated through experimental testing for motion predictions and Zemax OpticStudio (ZOS) simulations for optical link performance.

[0026] FIG. 1(a) presents the complete system architecture of the adaptive laser communication on a UAV platform. The framework operates through a combination of onboard processing on a Raspberry Pi 5 module and external computational offloading for DRL updates. The UAV's onboard IMU and navigation system records translational and rotational motion variables $\{x, y, z, \phi, \vartheta, \psi\}$, which are the UAV's 3D location coordinates (x, y, z) and orientation angles of pitch, yaw, and roll. Each measurement is time-stamped and transmitted to the onboard processor at a sampling frequency of 10 Hz. This data is streamed to the Raspberry Pi 5 for feature extraction and temporal prediction using a pre-trained neural network, which is updated every minute by uploading the latest IMU dataset to a computational server and downloading the newly trained weights and biases. The neural model, trained via a DDQN, estimates future

UAV motion states that are subsequently translated by the onboard controller into angular correction commands for the optical beam-steering system.

[0027] To ensure that motion forecasts remain temporally aligned with the UAV's actual dynamics, a prediction horizon Δt_p is selected to exceed the total latency of the sensing, processing, and control pipeline, including IMU data extraction and preprocessing time (t_{ext}), inference delay (t_{ML}), communication latency (t_{comm}), and actuator response time (t_{act}):

$$\Delta t_p \geq t_{ext} + t_{ML} + t_{comm} + t_{act}. \quad (1)$$

[0028] For the current system implementation, representative delays of t_{ext} = 100 ms, t_{ML} = 200 ms, t_{comm} = 10 ms, and t_{act} = 200 ms are adopted, resulting in a total latency of approximately 500 ms. These values are conservatively chosen to accommodate potential variations or transient delays across different hardware instances and operating conditions.

[0029] The optical subsystem consists of a small form-factor pluggable (SFP) transceiver, a laser collimator, and a beam expander, producing a well-collimated optical beam. The beam-steering assembly integrates a three-axis gimbal for coarse motion control and a MEMS mirror for fine angular alignment. The control unit computes correction angles from the predicted motion sequence, driving the MEMS mirror to predictively counteract UAV-induced jitter.

[0030] The computational offloading module operates on a longer timescale training loop, executing every minute. During each cycle, the most recent one minute of IMU telemetry is transmitted to an external computing node for multi-episode DDQN training. The resulting updated network weights are then uploaded to the onboard Raspberry Pi for deployment. This periodic learning process balances real-time latency constraints with offline training complexity, enabling continuous adaptation and stable optical beam correction throughout UAV flight. FIG. 1(b) shows the corresponding setup of the architecture. The complete optical stack, including mechanical housing and alignment tolerances, occupies an estimated volume of 300 mm × 200 mm × 200 mm. This architecture delivers high optical performance within a small-footprint design, making it particularly suitable for UAV-based FSO links where payload capacity and vibration resilience are critical.

[0031] FIG. 2 illustrates the DDQN framework used for predictive UAV motion estimation and beam control. The approach integrates temporal state sampling, experience replay, and dual-network learning to minimize overestimation bias and ensure convergence stability. At each episode, the UAV provides normalized IMU-derived states s_n sampled at 10 Hz over an observation window of $T_{\mathrm{obs}} = 3$ s. For each state, the DDQN agent selects an action a_n representing a corrective displacement in the predicted trajectory. Each tuple $\langle s_n, a_n, r_n, s'_n, d_n \rangle$ - containing the reward, next state, and done flag - is stored in a replay buffer for training. Each state s_n is formed from normalized positions $\{x(t), y(t), z(t)\}$ over the past T_{obs} s. For each axis, values are standardized using

$$Z = \frac{x-\mu}{\sigma} \tag{2}$$

[0032] where μ and σ are the mean and standard deviation. This normalization provides scale invariance and stability for training. The resulting z-scores for each axis, typically within $[-3, +3]$, constitute a 3D continuous state space in temporal order.

[0033] The action space consists of a corrective increment discretized into 101 bins within $\Delta Z \in [-0.2, 0.2]$, representing the predicted change relative to a position estimate, Z_t, extrapolated from the last two positions. This increment is added to Z_t and then converted back to the physical domain as

$$\hat{x}_{t+\Delta t} = (Z_t + \Delta Z)\sigma + \mu, \tag{3}$$

[0034] yielding the predicted UAV displacement, which is then mapped to a MEMS mirror control angle for beam steering. A smoothness penalty ($\lambda = 0.01$) discourages abrupt corrections. The Q-network approximates the state–action value function $Q(s, a; \theta)$ via two fully connected hidden layers, each containing 128 neurons with ReLU activation, while a target network with parameters θ_{targ} provides delayed target estimates. The Bellman target for the DDQN update is defined as

$$P = r + \gamma Q(s', a^{\max}; \theta_{\text{targ}}), \quad (4)$$

[0035] where the greedy action is determined by

$$a^{\max} = \arg\max_{a} Q(s', a; \theta) \quad (5)$$

[0036] and $\gamma = 0.99$ is the discount factor. The mean-squared error (MSE) loss between the predicted and target Q-values is:

$$L = \mathbb{E}\left[\left(P - Q(s, a; \theta)\right)^2\right] \quad (6)$$

[0037] Soft target updates are applied at every iteration using

$$\theta_{\text{targ}} \leftarrow \tau\theta + (1 - \tau)\theta_{\text{targ}}, \quad (7)$$

[0038] with $\tau = 10^{-3}$. The parameters are optimized using Adam with a learning rate of 10^{-3}, and the exploration rate ϵ decays gradually from 1.0 to 0.01 to ensure a smooth transition from exploration to exploitation. Training is conducted with mini-batches of 128 samples drawn from a replay memory of 2×10^5 tuples. The complete DDQN training and deployment workflow for UAV motion prediction and MEMS-based beam alignment is summarized in Algorithm 1.

Algorithm 1 DDQN Model for UAV Motion Prediction

Input: Replay buffer size M, batch size B, learning rate α, discount factor γ, target smoothing factor $\tau = 10^{-3}$, exploration rate ϵ, decay $\Delta\epsilon$
Initialize Q-network with parameters θ
Initialize target network with $\theta_{\text{targ}} \leftarrow \theta$
Initialize replay memory $\mathcal{D}$ with capacity M
for each training episode $k = 1, 2, \ldots, K$ **do**
    Reset environment and obtain initial UAV state s_1
    for each sampling step $n = 1, 2, \ldots, N$ **do**
        With probability ϵ, select a random action a_n
        Otherwise, select $a_n = \arg\max_a Q(s_n, a; \theta)$
        Execute a_n in the environment and observe s_{n+1}, r_n, and done flag d_n
        Store tuple $\langle s_n, a_n, r_n, s_{n+1}, d_n \rangle$ in $\mathcal{D}$
        if $|\mathcal{D}| \geq B$ **then**
            Sample random mini-batch $\mathcal{B}$ of B tuples from $\mathcal{D}$
            for each $(s_i, a_i, r_i, s'_i, d_i)$ in $\mathcal{B}$ **do**
                $a^{\max} \leftarrow \arg\max_a Q(s'_i, a; \theta)$ {Action selection (online net)}
                $P_i \leftarrow r_i + (1 - d_i)\gamma Q(s'_i, a^{\max}; \theta_{\text{targ}})$
            end for
            Compute loss $L = \frac{1}{B} \sum_i (P_i - Q(s_i, a_i; \theta))^2$
            Update parameters $\theta \leftarrow \theta - \alpha \nabla_\theta L$ {Adam optimizer}
            Update target net $\theta_{\text{targ}} \leftarrow \tau\theta + (1 - \tau)\theta_{\text{targ}}$
        end if
        Decay exploration: $\epsilon \leftarrow \max(\epsilon_{\min}, \epsilon - \Delta\epsilon)$
        if d_n is True **then**
            break {End of episode}
        end if
    end for
end for
Deployment: Use trained network to predict future UAV states and generate predictive MEMS beam-steering commands.

[0039] A full-fledged model of the optical system in ZOS was developed and the performance of the proposed DDQN-based SDO framework was evaluated via ray tracing simulations. A series of test flights on an industrial drone ARRIS M1200 under two motion conditions were performed: hovering and various translational motion. During each flight, IMU records accelerometer and gyroscope data at 10 Hz, capturing realistic vibration and drift patterns representative of UAV-induced beam misalignment. The hovering dataset is approximately 1250 sec long and the motion dataset is approximately 600 sec long; in both cases, the first half was used for training, and the remaining half was used for testing. These datasets are later extracted and used to drive the simulated transmitter motion in ZOS, allowing the optical model to replicate real-world UAV dynamics during beam propagation and coupling analysis.

[0040] A high-fidelity optical model was developed in Zemax OpticStudio (ZOS) 2022 R2.02 to simulate the laser communication link under realistic UAV-induced perturbations. ZOS enables accurate wavelength-dependent propagation, aperture diffraction, and alignment-error analysis. Real component parameters from Thorlabs were imported, including a TC18APC-1310 collimator, a Galilean beam expander (GBE05-C), and a MEMS steering mirror, to ensure consistency with the physical setup. Using the ZOS API, the model was automated to integrate UAV telemetry and perform multi-configuration ray tracing for multiple tilt and shift cases. UAV-induced perturbations were introduced via Coordinate Break surfaces that emulate translational offsets extracted from IMU vibration data, while angular deviations were excluded due to gimbal stabilization. Each simulation produced receiver-plane footprints and spatial power maps used to evaluate beam divergence and alignment sensitivity. FIG. 1(c) illustrates the modeled system layout and representative beam footprint variations under dynamic UAV motion.

[0041] FIG. 3 illustrates the spatial configuration of the experimental setup used for evaluating the UAV-based laser communication link. The system consists of a UAV-mounted optical transmitter performing real-time beam correction toward a stationary receiver positioned at 8 distinct spatial coordinates. Two receiver groups are defined: four at ground-station level (denoted in violet) and four at the same altitude as the UAV (denoted in red). The horizontal and vertical separation between adjacent receivers is set to d = 100 m, forming a 2 × 2 grid pattern at each elevation plane. The coordinate transformation between the UAV body frame (x, y, z) and the optical link frame (α, β, γ) is also shown. At the initial time t = 0, the UAV's position is defined as the origin (0, 0, 0) in the (x, y, z) reference frame. The α axis extends from the UAV's transmitter origin toward the target receiver, representing the optical propagation vector. The β and γ axes form an orthogonal plane corresponding to the receiver surface, which serves as the reference for tracking beam footprint displacement. This transformation is repeated for each of the eight test cases, corresponding to unique receiver locations within the defined spatial grid. During all tests, the receivers remain stationary while the UAV adjusts its optical beam based on DDQN-predicted motion states. This setup enables systematic evaluation of beam stabilization and alignment robustness across varying spatial orientations and link geometries.

[0042] FIG. 4 illustrates the prediction performance of the proposed DDQN-based model under two UAV operating regimes: hovering and translational motion. The results are benchmarked against a baseline case, where beam alignment is corrected solely using the most recent IMU signal. This baseline represents a reactive correction strategy that adjusts the optical axis based on instantaneous attitude measurements rather than forecasting future states.

[0043] In FIG. 4(a) the predicted beam footprint trajectories in *X*, *Y*, and *Z* closely follow their corresponding ground-truth signals, demonstrating high temporal consistency. The error profiles in FIG. 4(b) show mean absolute errors (MAEs) of 5.34 mm, 6.06 mm, and 8.78 mm for the *X*, *Y*, and *Z* axes, respectively, compared to 13.71 mm, 15.14 mm, and 17.15 mm from the IMU-only baseline. This more-than-twofold reduction confirms the DDQN's ability to anticipate fine-scale positional deviations, maintaining sub-centimeter accuracy during stationary hovering.

[0044] When the UAV transitions into translational motion, beam fluctuations intensify due to gimbal lag, vibration, and abrupt directional changes. As shown in FIG. 4(d) the DDQN-predicted footprints track the true trajectories across multiple flight phases, including lateral, longitudinal, and vertical transitions, while preserving smooth dynamics. The corresponding prediction errors in FIG. 4(e) report MAE values of 88.67 mm, 72.96 mm, and 18.93 mm for *X*, *Y*, and *Z*, respectively. These values significantly outperform the IMU-only baseline, which recorded 381.39 mm, 298.92 mm, and 29.37 mm. Despite the larger motion amplitude and higher-frequency disturbances, the DDQN maintains coherent trajectory prediction and predictive smoothness during UAV's translational motion, enabling predictive beam-steering adjustments under dynamic UAV flight.

[0045] FIG. 4(c) and FIG. 4(f) illustrate the beam footprint behavior on the receiver plane under two UAV operating regimes: hovering and translational motion. The comparison highlights the performance of the proposed DDQN-based predictive correction framework against the baseline beam alignment method.

[0046] Under hovering, the beam footprints exhibit a concentrated distribution near the receiver center, indicating effective stabilization in low-dynamic conditions. A 3D density map in FIG. 4(c) quantifies containment within $R = 10$ mm and $R = 20$ mm zones, where the DDQN controller confines 60.1% and 93.5% of footprints, respectively, compared to 22.1% and 60.0% for the baseline. These results confirm that the DDQN controller mitigates residual micro-vibrations and orientation drifts inherent in UAV hovering.

[0047] During translational motion, the beam footprint spread increases significantly due to higher dynamic disturbances, including gimbal lag, translational vibration, and angular jitter. The density map in FIG. 4(f) shows that the DDQN model confines 30.3% and 66.2% of footprints within $R = 25$ mm and $R = 75$ mm, respectively, compared to 9.7% and 30.2% under the baseline. Despite the higher motion dynamics, the DDQN framework maintains robust predictive stabilization and improved optical link consistency during UAV's translational motion.

[0048] FIG. 5 presents the coupling efficiency trends for varying prediction horizons. While $T_{\text{pred}} = 0.5$ s serves as the standard configuration, additional cases with longer horizons are included to examine how extending the prediction window affects optical link stability and the potential for onboard model retraining.

[0049] In FIG. 5(a), corresponding to the hovering condition and for $T_{\text{pred}} = 0.5$ s case, DDQN-corrected response outperforms the baseline, achieving near-unity coupling at substantially smaller receiver apertures. The $T_{\text{pred}} = 0.5$ s case exhibits the steepest rise, indicating the most accurate short-term compensation, whereas the 0.75 s and 0.9 s horizons does not produce results better than the baseline. The inset plot in logarithmic scale highlights this difference.

[0050] FIG. 5(b) shows the results for UAV translational motion, where beam wander and dynamic misalignment are more pronounced. The DDQN-corrected responses outperform the baseline IMU-only approach. Although longer prediction horizons provide additional processing time that could be utilized for adaptive retraining onboard, they also introduce a gradual loss of efficiency relative to the 0.5 s case. This trade-off illustrates the balance between prediction latency and real-time accuracy: shorter horizons offer sharper beam confinement, while longer ones allow limited learning adaptation at the expense of slight coupling degradation.

[0051] Overall, these results confirm that the DDQN framework sustains high optical power capture across both static and dynamic UAV conditions. The 0.5 s prediction horizon remains optimal for low-latency beam correction, whereas modestly longer horizons may be leveraged for hybrid real-time and adaptive training without severe performance penalties.

[0052] The proposed DDQN-based beam correction framework demonstrates strong potential for maintaining optical link stability during UAV motion; however, several research directions remain to enhance its scalability, adaptability, and deployment readiness. Future work can extend the current single-UAV control model toward multi-agent reinforcement learning (MARL), enabling cooperative beam steering among multiple drones that share link-state data to improve spatial coverage, link continuity, and fault tolerance in swarm-based laser communication networks. Real-time inference on embedded platforms such as the Raspberry Pi 5 can be further optimized through hardware acceleration using FPGAs, TPUs, or edge AI chips, while pruning and quantization techniques can minimize latency and power consumption for onboard implementation. Environmental adaptability may also be achieved by incorporating turbulence intensity, weather parameters, or optical feedback into the DDQN policy, allowing the system to dynamically respond to refractive-index fluctuations and maintain robust performance in outdoor propagation. Additionally, integrating IMU-based prediction with vision or LIDAR feedback could enable hybrid control architectures that compensate both rapid jitter and slow positional drift through hierarchical sensor fusion. Finally, extended-range outdoor testing beyond 250 m, involving moving receivers and high-altitude UAV relays, will be essential to validate the framework's stability under turbulence, occlusion, and real atmospheric conditions. Collectively, these advancements will transform the presented DDQN-based controller into a fully autonomous, scalable, and weather-resilient architecture for next-generation aerial laser communication networks.

[0053] The present invention provides a DRL-driven beam-steering framework that leverages a DDQN to predict UAV motion and perform predictive optical corrections in real time. Using onboard IMU data, the model forecasts positional deviations and translates them into angular compensation commands for a MEMS-based beam steering system. The integrated setup, comprising a Raspberry Pi 5 controller, MEMS mirror, and compact optical transmitter, was validated under both hovering and motion conditions using eight receiver positions. The proposed system achieved substantial improvement in optical coupling efficiency compared to a baseline non-predictive case. By demonstrating the feasibility of predictive optical alignment using lightweight onboard hardware, this work establishes a foundation for autonomous, low-latency optical links between aerial platforms. The framework can be readily extended to multi-agent coordination, adaptive turbulence compensation, and long-range outdoor operation, paving the way toward intelligent, self-stabilizing optical networks for next-generation UAV communication systems.

[0054] Various embodiments of the present invention may be embodied on a computer readable medium. Program code embodied on a computer readable medium may be transmitted using any appropriate medium, including but not limited to wireless, wire-line, optical fiber cable, radio frequency, etc., or any suitable combination of the foregoing. Computer program code for carrying out operations for aspects of the present invention may be written in any combination of one or more programming languages, including an object-oriented programming language such as Java, C#, C++, Visual Basic or the like and conventional procedural programming languages, such as the "C" programming language or similar programming languages.

[0055] Aspects of the present invention are described below with reference to flowchart illustrations and/or block diagrams of methods, apparatus (systems) and computer program products according to embodiments of the invention. It will be understood that each block of the flowchart illustrations and/or block diagrams, and combinations of blocks in the flowchart illustrations and/or block diagrams, can be implemented by computer program instructions. These computer program instructions may be provided to a processor of a general-purpose computer, special purpose computer, or other programmable data processing apparatus to produce a machine, such that the instructions, which execute via the processor of the computer or other programmable data processing apparatus, create means for implementing the functions/acts specified in the flowchart and/or block diagram block or blocks.

[0056] These computer program instructions may also be stored in a computer readable medium that can direct a computer, other programmable data processing apparatus, or other devices to function in a particular manner, such that the instructions stored in the computer readable medium produce an article of manufacture including instructions which implement the function/act specified in the flowchart and/or block diagram block or blocks.

[0057] The computer program instructions may also be loaded onto a computer, other programmable data processing apparatus, or other devices to cause a series of operational steps to be performed on the computer, other programmable apparatus or other devices to produce a computer implemented process such that the instructions which execute on the computer or other programmable apparatus provide processes for implementing the functions/acts specified in the flowchart and/or block diagram block or blocks.

[0058] The advantages set forth above, and those made apparent from the foregoing description, are efficiently attained. Since certain changes may be made in the above construction without departing from the scope of the invention, it is intended that all matters contained in the foregoing description or shown in the accompanying drawings shall be interpreted as illustrative and not in a limiting sense.

[0059] It is also to be understood that the following claims are intended to cover all of the generic and specific features of the invention herein described, and all statements of the scope of the invention that, as a matter of language, might be said to fall therebetween.

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What is claimed is:

1. A method for predictive beam steering in an unmanned aerial vehicle (UAV) laser communication system, comprising:

   receiving inertial measurement unit (IMU) telemetry data from a UAV, the IMU telemetry data including position and motion state variables derived from accelerometer and gyroscope measurements;

   predicting, using a Double Deep Q-Network (DDQN), future UAV motion states based on the IMU telemetry data;

   generating predictive beam-steering commands from the predicted future UAV motion states;

   actuating a micro-electro-mechanical system (MEMS) mirror at a laser transmitter on the UAV based on the predictive beam-steering commands to compensate for motion-induced misalignment; and

   transmitting a laser beam aligned by the MEMS mirror to maintain optical link performance.

2. The method of claim 1, wherein the IMU telemetry data includes motion state variables derived from accelerometer and gyroscope measurements, the motion state variables including estimated three-dimensional position coordinates and rotational motion parameters corresponding to roll, pitch, and yaw.

3. The method of claim 1, wherein predicting the future UAV motion states includes selecting a prediction horizon that exceeds a total latency of sensing, processing, communication, and actuation.

4. The method of claim 1, wherein generating the predictive beam-steering commands includes translating the predicted future UAV motion states into angular correction commands for the MEMS mirror.

5. The method of claim 1, further comprising periodically updating the DDQN by offloading recent IMU telemetry to an external server for training and downloading updated network weights to an onboard processor.

6. A system for predictive beam steering in UAV laser communication, comprising: an inertial measurement unit (IMU) on a UAV configured to provide telemetry data including position and rotational motion variables;

   a processor configured to execute a Double Deep Q-Network (DDQN) to predict future UAV motion states from the IMU telemetry data and generate predictive beam-steering commands;

   a micro-electro-mechanical system (MEMS) mirror at a laser transmitter on the UAV;

   a control unit configured to actuate the MEMS mirror based on the predictive beam-steering commands to align a transmitted laser beam; and

   an optical subsystem including a transceiver, collimator, and beam expander for producing the laser beam.

7. The system of claim 6, further comprising a computational offloading module configured to transmit recent IMU telemetry to an external server for DDQN training and receive updated network weights.

8. The system of claim 6, wherein the control unit combines the predicted future UAV motion states with a 3D gimbal for coarse control and the MEMS mirror for fine alignment.

9. A non-transitory computer-readable medium storing instructions that, when executed by a processor, cause the processor to:

   receive inertial measurement unit (IMU) telemetry data from a UAV, including position and rotational motion variables;

   predict, via a Double Deep Q-Network (DDQN), future UAV motion states based on the IMU telemetry data;

generate predictive beam-steering commands from the predicted future UAV motion states;

command actuation of a micro-electro-mechanical system (MEMS) mirror at a laser transmitter on the UAV using the predictive beam-steering commands to align a laser beam; and

periodically update the DDQN by offloading IMU telemetry to an external server for training.

10. The non-transitory computer-readable medium of claim 9, wherein the instructions further cause the processor to select a prediction horizon exceeding total system latency.

11. The non-transitory computer-readable medium of claim 9, wherein the predictive beam-steering commands compensate for UAV vibrations and turbulence.

12. The non-transitory computer-readable medium of claim 9, wherein the DDQN training uses multi-episode reinforcement learning on one-minute IMU datasets.

ABSTRACT

A Software-Defined Optics (SDO) framework that brings programmability, prediction, and intelligent control to UAV FSOC communication is provided. Instead of relying solely on reactive feedback, the SDO framework uses onboard inertial data and machine learning to predict UAV motion before it causes optical drift. A reinforcement-learning-based controller predicts how the UAV will move and proactively adjusts a beam-steering module to maintain alignment. By transforming the optical layer into a software-defined, learning-enabled system, the SDO approach enables FSOC links to withstand rapid motion that would normally break communication.

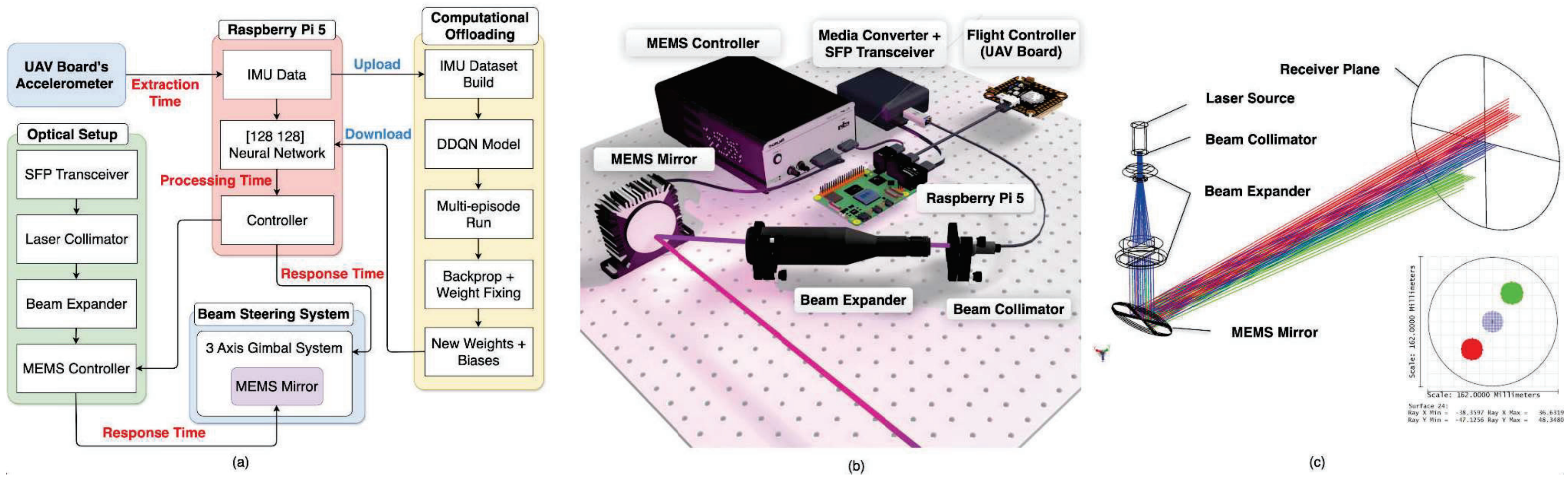
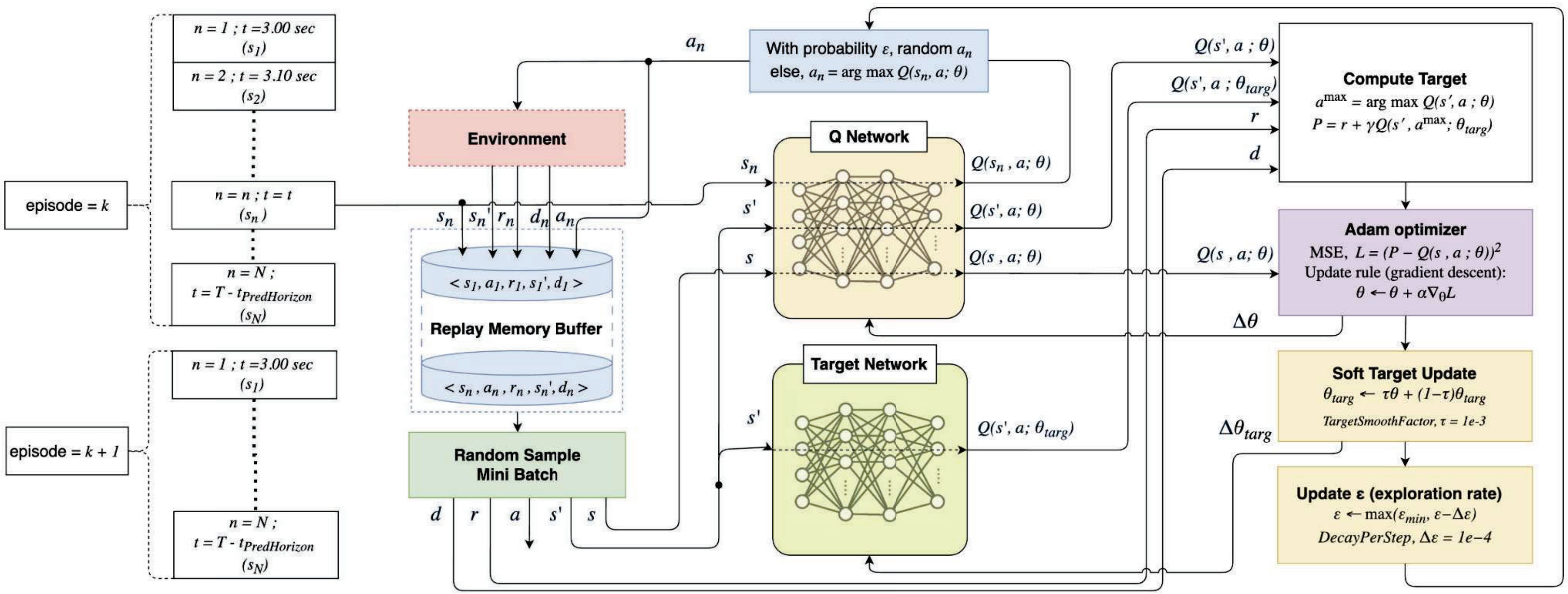

UAV Board's Accelerometer
Extraction Time
Raspberry Pi 5
IMU Data
[128 128] Neural Network
Processing Time
Controller
Upload
Download
Computational Offloading
IMU Dataset Build
DDQN Model
Multi-episode Run
Backprop + Weight Fixing
New Weights + Biases
Optical Setup
SFP Transceiver
Laser Collimator
Beam Expander
MEMS Controller
Response Time
Beam Steering System
3 Axis Gimbal System
MEMS Mirror
Response Time
(a)
MEMS Controller
Media Converter + SFP Transceiver
Flight Controller (UAV Board)
MEMS Mirror
Raspberry Pi 5
Beam Expander
Beam Collimator
(b)
Receiver Plane
Laser Source
Beam Collimator
Beam Expander
MEMS Mirror
(c)

Fig. 1



episode = k
$n = 1$; $t = 3.00$ sec (s_1)
$n = 2$; $t = 3.10$ sec (s_2)
$n = n$; $t = t$ (s_n)
$n = N$; $t = T - t_{PredHorizon}$ (s_N)
episode = $k + 1$
$n = 1$; $t = 3.00$ sec (s_1)
$n = N$; $t = T - t_{PredHorizon}$ (s_N)
Environment
s_n s_n' r_n d_n a_n
< s_1, a_1, r_1, s_1', d_1 >
Replay Memory Buffer
< s_n, a_n, r_n, s_n', d_n >
Random Sample Mini Batch
d r a s' s
a_n
With probability ε, random a_n else, $a_n = \arg\max Q(s_n, a; \theta)$
Q Network
s_n s' s
$Q(s_n, a; \theta)$
$Q(s', a; \theta)$
$Q(s, a; \theta)$
Target Network
s'
$Q(s', a; \theta_{targ})$
Compute Target
$a^{max} = \arg\max Q(s', a; \theta)$
$P = r + \gamma Q(s', a^{max}; \theta_{targ})$
$Q(s', a; \theta)$
$Q(s', a; \theta_{targ})$
r
d
Adam optimizer
MSE, $L = (P - Q(s, a; \theta))^2$
Update rule (gradient descent):
$\theta \leftarrow \theta + \alpha \nabla_\theta L$
$Q(s, a; \theta)$
$\Delta\theta$
Soft Target Update
$\theta_{targ} \leftarrow \tau\theta + (1-\tau)\theta_{targ}$
TargetSmoothFactor, $\tau = 1e\text{-}3$
$\Delta\theta_{targ}$
Update ε (exploration rate)
$\varepsilon \leftarrow \max(\varepsilon_{min}, \varepsilon - \Delta\varepsilon)$
DecayPerStep, $\Delta\varepsilon = 1e{-4}$

Fig. 2

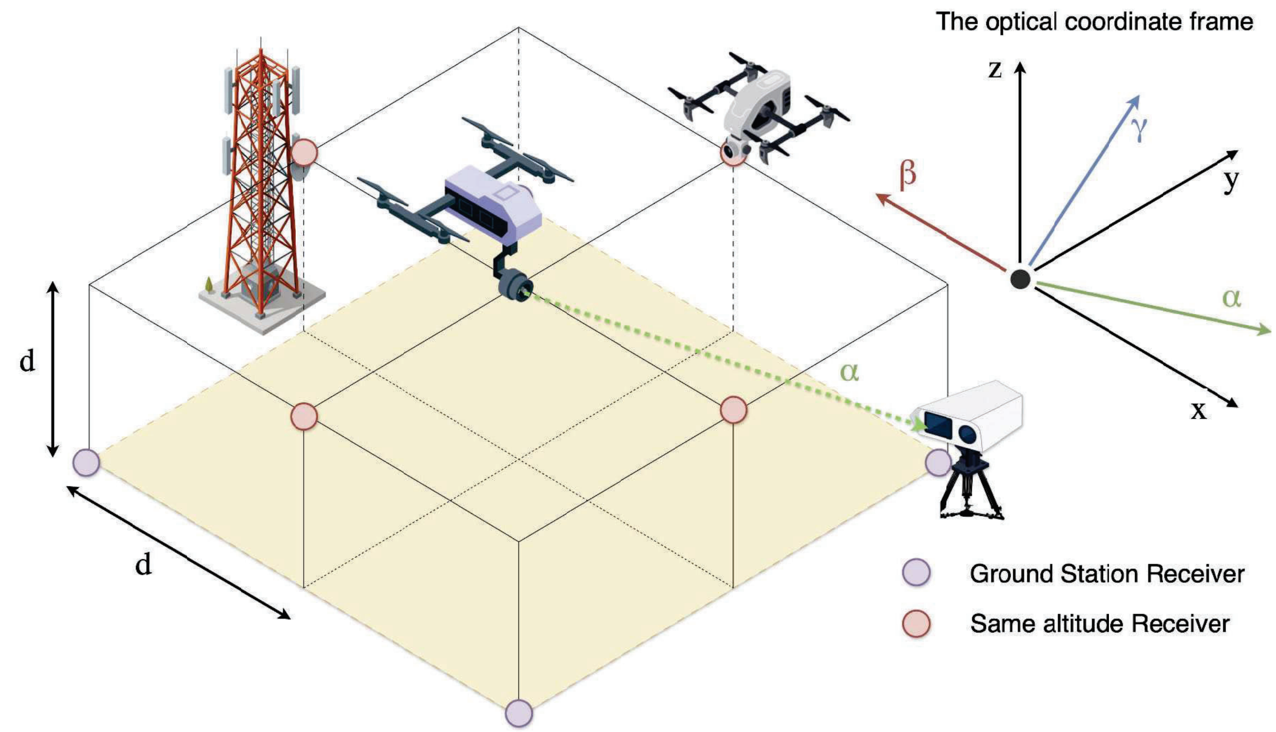

The optical coordinate frame
z
γ
y
β
α
x
d
d
α
Ground Station Receiver
Same altitude Receiver

Fig. 3

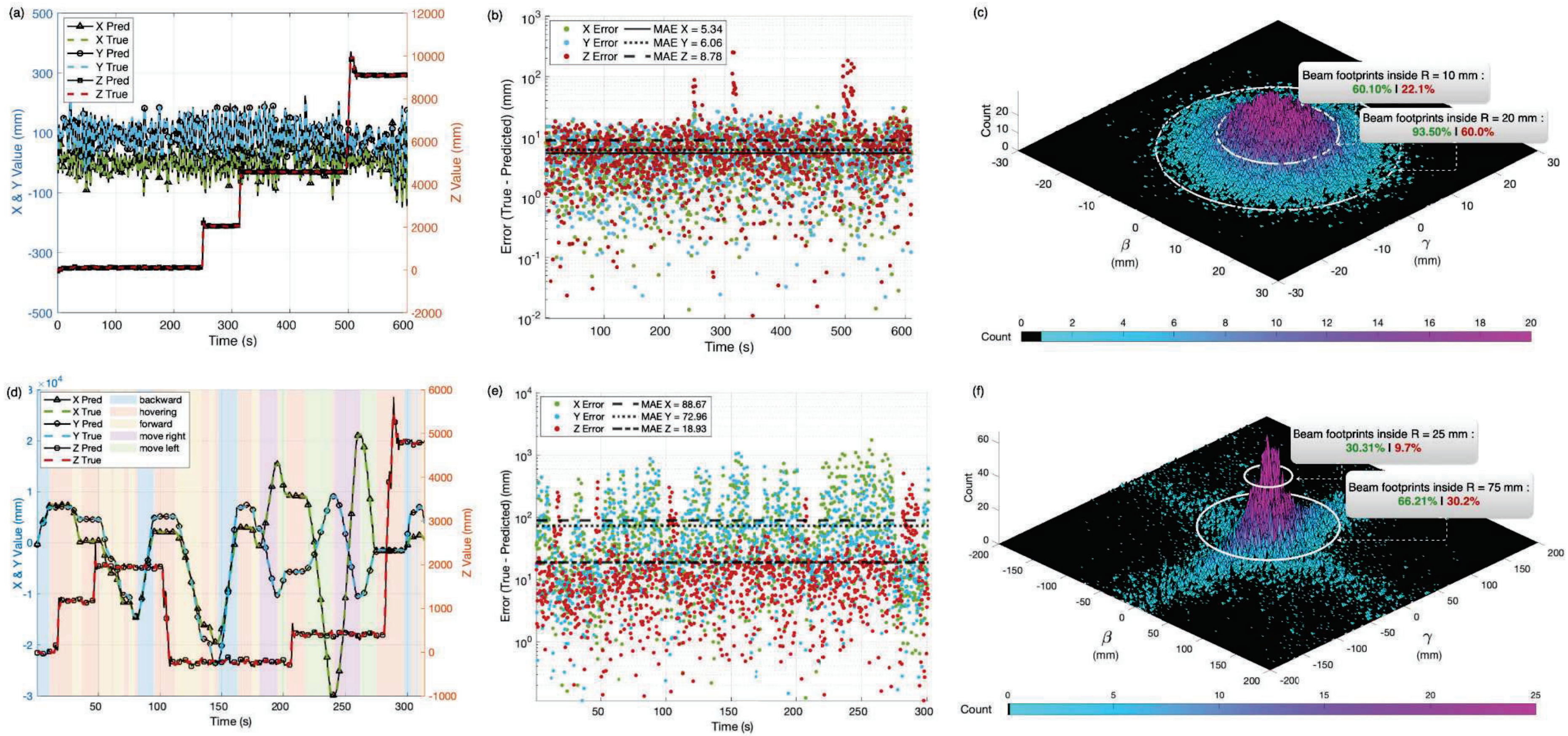

(a)
X Pred
X True
Y Pred
Y True
Z Pred
Z True
X & Y Value (mm)
Z Value (mm)
Time (s)
(b)
X Error
Y Error
Z Error
MAE X = 5.34
MAE Y = 6.06
MAE Z = 8.78
Error (True - Predicted) (mm)
Time (s)
(c)
Beam footprints inside R = 10 mm :
60.10% | 22.1%
Beam footprints inside R = 20 mm :
93.50% | 60.0%
Count
β (mm)
γ (mm)
(d)
X Pred
X True
Y Pred
Y True
Z Pred
Z True
backward
hovering
forward
move right
move left
X & Y Value (mm)
Z Value (mm)
Time (s)
(e)
X Error
Y Error
Z Error
MAE X = 88.67
MAE Y = 72.96
MAE Z = 18.93
Error (True - Predicted) (mm)
Time (s)
(f)
Beam footprints inside R = 25 mm :
30.31% | 9.7%
Beam footprints inside R = 75 mm :
66.21% | 30.2%
Count
β (mm)
γ (mm)

Fig. 4

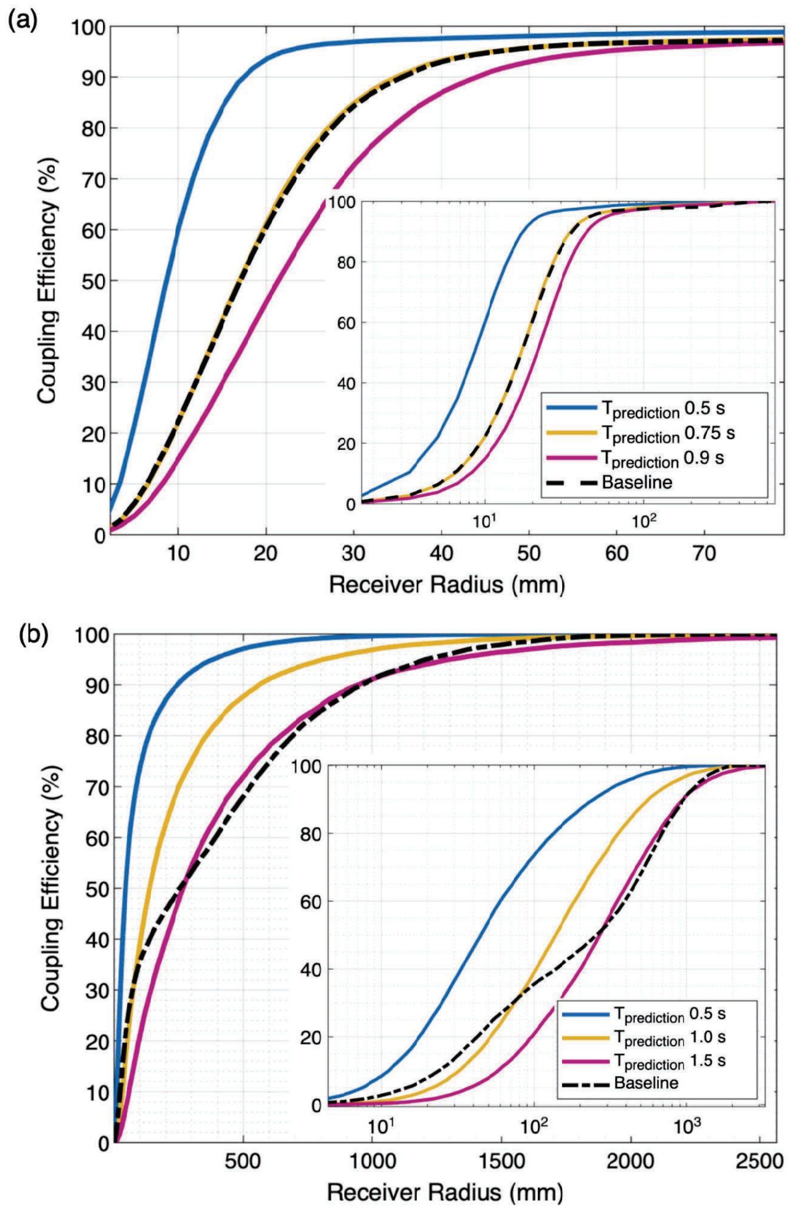

(a)
Coupling Efficiency (%)
Receiver Radius (mm)
T prediction 0.5 s
T prediction 0.75 s
T prediction 0.9 s
Baseline
(b)
Coupling Efficiency (%)
Receiver Radius (mm)
T prediction 0.5 s
T prediction 1.0 s
T prediction 1.5 s
Baseline

Fig. 5